Shamir Khan

Ontario, Canada | **(647)-606-6063** | shamirk2121@gmail.com | linkedin.com/in/shamir-khan-40698123b | github.com/shamir-khan-dev | shamir-khan-dev.github.io

Education

Bachelor of Science in Computer Science **Expected April 2028**
Wilfrid Laurier University *Waterloo, ON*

Relevant Coursework: Object-Oriented Programming (OOP), Data Structures & Algorithms, Distributed Systems, Operating Systems, Database Systems, Computer Networks, Systems Architecture, Discrete Mathematics

Experience

AI Software Engineer (Client Project) **Oct 2026 – Present**
Go Fig Bookz (via Riipen Labs) *Remote, ON*

- **Architected an autonomous AI bookkeeping agent** in **TypeScript** and **Node.js** to automate invoice reconciliation, achieving a **100% tie-out** with **$0.00 discrepancy** across $17K+ in client disbursements.
- **Engineered three-way document matching algorithms** and automated HST tax auditing, detecting duplicate vendor invoices and reducing manual accounting reconciliation time by **80%**.
- **Authored comprehensive Vitest test suites** covering 43 automated tests across 9 suites, validating multi-bank statement parsers, deterministic ledger writes, and zero-regression audit pipelines.

Lead Web Developer & Systems Administrator **Jan 2026 – Present**
Wilfrid Laurier University Student Association (WLU MSA) *Waterloo, ON*

- **Architected and deployed a production web platform** serving **250+ active campus members** using **Next.js**, **React**, **TypeScript**, and **Tailwind CSS**, maintaining **99.9% uptime** (wlumsa.org).
- **Engineered dynamic donation and event modules** integrated with **Payload CMS** and payment webhooks, supporting student-led initiatives that processed **$50K+** in charitable contributions.

Projects

CloudGuard – Cloud Security Posture Management (CSPM) | *Live Site* | *GitHub* **July 2026**

- **Architected a real-time cloud security platform** across 5 containerized microservices connected via **Apache Kafka**, monitoring cloud infrastructure telemetry across AWS, Azure, and GCP with **99.9% uptime**.
- **Built a native C++ telemetry agent** using Windows kernel APIs (`GlobalMemoryStatusEx`, `GetSystemTimes`) and POSIX parsers to stream machine metrics to Kafka topics at sub-millisecond overhead.
- **Developed an unsupervised scikit-learn IsolationForest ML pipeline** scoring live Kafka streams in **507ms (p50)**, detecting SSH brute-force intrusions and billing spikes to cut incident response MTTR by **85%**.

NexusQueue – Distributed Asynchronous Task & DAG Engine | *Live Site* | *GitHub* **Nov 2026**

- **Engineered an asynchronous background job processing engine** (benchmarked at **119,000 jobs/sec**) to offload compute-heavy tasks from web servers across 50 concurrent workers with zero task loss.
- **Implemented a Binary Min-Heap Priority Queue** with $O(\log n)$ dispatch, ensuring high-priority jobs execute first; combined with **Token Bucket** limiters and AWS Full Jitter backoff against thundering herds.
- **Designed a DAG Workflow Scheduler** executing multi-stage dependent tasks via **Kahn's topological sort**, paired with an automated **Zombie Worker Watchdog** and live WebSocket telemetry dashboard.

LedgerMesh – High-Concurrency Double-Entry Ledger Core | *Live Site* | *GitHub* **Nov 2026**

- **Built a high-throughput financial transaction ledger** (benchmarked at **10,790 tx/sec**, **0.05ms** latency) enforcing double-entry bookkeeping (**Total Debits = Total Credits**) to prevent double-spending and floating-point drift.
- **Designed Optimistic Concurrency Control (OCC)** with account versioning and backoff, eliminating database write locks across 20 worker coroutines; integrated **SHA-256 Merkle audit trails** and Prometheus metrics.

Technical Skills Summary

Languages: C, C++, Python, Java, TypeScript, JavaScript, SQL, Scala, HTML5/CSS3
Distributed Systems & Cloud: Apache Kafka, Apache Spark, Redis Streams, Docker, WebSockets, GitHub Actions (CI/CD), Linux/Unix, AWS, Render, Vercel
Frameworks & Databases: FastAPI, Node.js, Express, React, Next.js, Spring Boot, PostgreSQL, MongoDB, SQLite, Vitest, NumPy, Pandas, OpenCV, Tailwind CSS
Core Concepts: AI Agents & Autonomous Workflows, LLM / Generative AI Integration, Prompt Design, Test-Driven Development (TDD), Software Development Life Cycle (SDLC), Optimistic Concurrency Control (OCC), Binary Min-Heaps, DAG Scheduling, SHA-256 Merkle Trees, RESTful Web Services, Real-Time Systems